

Vladimir LAPIN, Ph.D.

Senior Research Engineer | Applied Mathematician | HPC Specialist

PERSONAL DATA

NATIONALITY: Russian | Authorized to work in the EU (Permanent Resident – Spain)
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PROFILES: [Website](#) | [GitHub](#) | [LinkedIn](#) | [Google Scholar](#) | [ORCID](#)

PROFILE SUMMARY

- Scientific Software Engineer:** Applied Mathematician (1st Honours, MSU) with 15 years developing parallel Fortran/MPI ocean/climate solvers and owning production modeling pipelines from numerical schemes to petascale execution.
- HPC Performance & Numerics:** Identifies scaling bottlenecks from domain-decomposition first principles (stencils, halo exchange, memory bandwidth) to optimize multi-node ocean model performance.
- Production Infrastructure & AI Adoption:** Co-lead for Auto-EC-Earth4 ensemble workflows; leading group-wide AI adoption through Climate Commons (CVC-COMMONS), defining agent contracts and CI validation harnesses for automated scientific computing.

SKILLS

NUMERICS	PDE discretization (FD/FV, C-grid, staggered), parallel solver design, domain decomposition, halo exchange, numerical stability analysis
CODE	Python, C/C++, FORTRAN (MPI, OPENMP), GIT, YAML, JSON, MATLAB, COMSOL
DATA	Python (xarray, numpy, pandas), NETCDF, CDO, High-performance Data Pipelines
INFRA	Linux/BASH, HPC (SLURM, LMOD), Workflow Orchestration (AUTOSUBMIT), CI/CD, Containers (APPTAINER)
CONSORTIA	Co-maintainer of EC-Earth4 across 30 partner institutes; operational release coordination, integration standards, CI/CD, documentation
MENTORING	Supervision of 2 PhD and 3 Master’s students; organization of training events

PROFESSIONAL EXPERIENCE

2025 – now	Senior Research Engineer, Earth Sci. Dept., BSC, Spain
ROLE	◦ Co-Lead Engineer, Auto-EC-Earth4: Designing and operating the Autosubmit-based ensemble workflow layer for BSC’s CMIP7/WMO production runs on HPC clusters. ◦ AI Adoption & Modern Software Engineering: Leading AI adoption across the research group through Climate Commons (CVC-COMMONS); architecting institutional agent contracts (AGENTS.MD), GitHub/GitLab agentic workflows, and schema validation harnesses for automated scientific workflow ingestion and drift detection. ◦ Performance from Numerics: Identified MPI halo-exchange (lbc_lnk) as NEMO4 scalability bottleneck from domain-decomposition first principles; <code>nn_comm</code> switch delivered ~10% SYPD gain with no code changes.
2019 – 2024	Recognized Research Engineer, Earth Sci. Dept., BSC, Spain
ROLE	◦ Sea-ice Data Assimilation: Designed and implemented a nudging scheme for sea-ice state estimation in EC-Earth3; maintained production nudging pipelines for ocean and atmosphere components. ◦ Ensemble Infrastructure: Built and operated high-throughput ensemble systems (10s–100s of members, 50–100 SYPD) on HPC clusters for decadal prediction campaigns.

- 2015 – 2017 **Postdoctoral Scientist**, [Ocean Dept., MPI-M, Germany](#)
 ROLE ○ **Sea-ice Dynamics Solver**: Implemented an iterative EVP-class rheology solver for the ICON icosahedral-unstructured grid, where standard domain decomposition assumptions do not hold; addressed the poor parallel scalability inherent to sea-ice dynamics.
- 2011 – 2014 **Research Fellow**, [Applied Math Dept., Univ. of Leeds, UK](#)
 ROLE ○ **Novel FD Scheme**: Derived and implemented a novel second-order accurate C-grid boundary treatment eliminating $O(1)$ staircase errors at irregular coastlines (*Ocean Modelling*, 2014).
 ○ **Global Baroclinic Tide Model**: Built a global spectral-FD solver for coupled barotropic/baroclinic tides at 2 arc-minute resolution (8 constituents, Fortran/MPI); resolved internal tide generation over rough topography at scale.
- SUM 2008 **Risk Analyst (Intern)**, [Raiffeisenbank, Russia](#)
 ROLE ○ **Data Analysis**: Portfolio risk diagnostics, quantitative metric tracking, and preparation of operational risk reports.
- SUM 2007 **Junior Statistician**, [ACNielsen, Russia](#)
 ROLE ○ **Statistics**: Statistical analysis of retail consumer purchase data and anomaly detection in store audit datasets.

EDUCATION

- 2008 – 2011 **Ph.D.** in *Applied Mathematics*, [University of Limerick, Ireland](#)
 ROLE ○ **Applied Math**: Research in geophysical fluid dynamics and thin-film Stokes flows using asymptotic analysis and numerical modelling (MATLAB, COMSOL).
- 2003 – 2008 **B.Sc., M.Sc.** in *Applied Mathematics*, [Moscow State University, Russia](#)
 ○ **Grades**: 4.85/5*, 1st Honours

SELECTED PUBLICATIONS

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| 2024 | Silvy, Y et al. (Dec. 2024). “AERA-MIP: emission pathways, remaining budgets, and carbon cycle dynamics compatible with 1.5 and 2 °C global warming stabilization”. <i>Earth System Dynamics</i> 15 , 1591–1628. ISSN: 2190-4987. |
| 2022 | Hermanson, L et al. (Apr. 2022). “WMO Global Annual to Decadal Climate Update: A Prediction for 2021–25”. <i>Bulletin of the American Meteorological Society</i> 103 , E1117–E1129. ISSN: 1520-0477. |
| 2020 | Tsujino, H et al. (Aug. 2020). “Evaluation of global ocean–sea-ice model simulations based on the experimental protocols of the Ocean Model Intercomparison Project phase 2 (OMIP-2)”. <i>Geoscientific Model Development</i> 13 , 3643–3708. ISSN: 1991-9603. |

Last updated: September 24, 2026